

CS1231S TUTORIAL #2

Predicate Logic and Proofs

(AY2025/26 Semester 1) – T11B

Adapted from Prof Aaron Tan's slides

Learning objectives for this tutorial:

- Knowing the **contrapositive**, **converse** and **inverse** forms of a conditional statement. (Q1)
- Understanding **quantified statements**. (Q5, Q6)
- **Writing quantified statements** using $\forall$ and $\exists$. (Q2, Q3, Q9, Q10)
- **Wrong** proofs. (Q7)
- Writing **proofs**. (Q4, Q8, Q11)

Quick Pointers

- Common mistakes:
 - Using commas instead of conjunction / disjunction / etc.
 - Predicates as 'functions': Loves(x, Reindeer(y)) is non-sensical!

$$\forall x \in D, P(x) \equiv \forall x ((x \in D) \rightarrow P(x))$$

$$\exists x \in D, P(x) \equiv \exists x ((x \in D) \wedge P(x))$$

$$\forall x (\exists y P(x, y) \vee Q(x, y)),$$

Q1 Converse, inverse and contrapositive.

Quick recap:

Conditional statement: $p \rightarrow q$

Converse: $q \rightarrow p$

Inverse: $\sim p \rightarrow \sim q$

Contrapositive: $\sim q \rightarrow \sim p$

Reminder: Converse is equivalent to the inverse

Q1 Converse, inverse and contrapositive.

Are they true or false?

$$(a) \forall n \in \mathbb{Z} (6 \mid n \rightarrow 2 \mid n \wedge 3 \mid n)$$

True

Converse: $\forall n \in \mathbb{Z} (2 \mid n \wedge 3 \mid n \rightarrow 6 \mid n)$ True

Inverse: $\forall n \in \mathbb{Z} (6 \nmid n \rightarrow 2 \nmid n \vee 3 \nmid n)$ True

Contrapositive: $\forall n \in \mathbb{Z} (2 \nmid n \vee 3 \nmid n \rightarrow 6 \nmid n)$ True

Q1 Converse, inverse and contrapositive.

Are they true or false?

$$(b) \forall x (x \in \mathbb{Q} \rightarrow x \in \mathbb{Z})$$

False

Converse: $\forall x (x \in \mathbb{Z} \rightarrow x \in \mathbb{Q})$

True

Inverse: $\forall x (x \notin \mathbb{Q} \rightarrow x \notin \mathbb{Z})$

True

Contrapositive: $\forall x (x \notin \mathbb{Z} \rightarrow x \notin \mathbb{Q})$

False

Counterexample for original statement and contrapositive?

$$x = 0.5, 1.5, \text{etc.}$$

Q1

Converse, inverse and contrapositive.

Are they true or false?

(c) $\forall p, q \in \mathbb{Z} (Even(p) \wedge Even(q) \rightarrow Even(p + q))$.

True

Converse: $\forall p, q \in \mathbb{Z} (Even(p + q) \rightarrow Even(p) \wedge Even(q))$.

False

Inverse: $\forall p, q \in \mathbb{Z} (\sim Even(p) \vee \sim Even(q) \rightarrow \sim Even(p + q))$.

False

Contrapositive: $\forall p, q \in \mathbb{Z} (\sim Even(p + q) \rightarrow \sim Even(p) \vee \sim Even(q))$.

True

Counterexample for converse and inverse?

$$p = 1, q = 3, \text{ etc.}$$

Q2 (a) There is no biggest number, i.e. no matter how big a number is, there is always another number that is bigger.

What about this?

$$\exists x \in \mathbb{N} \forall y \in \mathbb{N} (y > x)$$

Answer: $\forall x \in \mathbb{N} \exists y \in \mathbb{N} (y > x)$

Works for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ but not $\mathbb{C}$.

Slightly clearer sentence form:

For all numbers, there is a number that is bigger than said number.



Q2 (b) Given any two distinct numbers, you can always find another number between them.

Answer:

$$\underline{\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} (x < y \rightarrow \exists z \in \mathbb{Q} (x < z) \wedge (z < y))}.$$

May shorten to $\forall x, y \in \mathbb{Q}$

Works for $\mathbb{Q}, \mathbb{R}$ but not $\mathbb{Z}, \mathbb{C}$.

Observe that we need to specify $x < y$. If not, our statement is false.



Q3

- (a) R is **reflexive** if and only if “ xRx for any x in A ”.
- (b) R is **symmetric** if and only if “for every x and y in A , if xRy then yRx ”.
- (c) R is **transitive** if and only if “for all x, y and z in A , if xRy and yRz , then xRz ”.

$$(a) \quad \forall x \in A (xRx)$$

$$(b) \quad \forall x, y \in A (xRy \rightarrow yRx)$$

$$(c) \quad \forall x, y, z \in A (xRy \wedge yRz \rightarrow xRz)$$

Sneak peak for later topics!

Q4 Prove or disprove:

- (a) Integers are closed under division.
- (b) Rational numbers are closed under addition.
- (c) Rational numbers are closed under division.

Counterexample for (a)?

Counterexample for (c)?

Q4 Prove: (b) Rational number are closed under addition.

Proof:

1. Let r and s be rational numbers.
2. Then $\exists a, b, c, d \in \mathbb{Z}$ s.t. $r = \frac{a}{b}, s = \frac{c}{d}, b \neq 0, d \neq 0$. (by definition of rational numbers)
3. Hence, $r + s = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$. (by basic algebra)
4. $ad + bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}$. (closure of $\mathbb{Z}$ under $+$ and $\times$)
5. Moreover, $bd \neq 0$ since $b \neq 0, d \neq 0$. (Appendix A T11: Zero Product Property)
6. Hence, $r + s$ is rational. (by definition of rational numbers)
7. Therefore, rational numbers are closed under addition.



$$r \text{ is rational} \Leftrightarrow \exists a, b \in \mathbb{Z} \text{ s.t. } r = \frac{a}{b} \text{ and } b \neq 0.$$

Appendix 1

While these are rarely explicitly tested in exam, they will be useful for scoring maximum marks in Assignments.

When in doubt, just include the law.

F1. *Commutative Laws* For all real numbers a and b ,

$$a + b = b + a \quad \text{and} \quad ab = ba.$$

F2. *Associative Laws* For all real numbers a , b , and c ,

$$(a + b) + c = a + (b + c) \quad \text{and} \quad (ab)c = a(bc).$$

F3. *Distributive Laws* For all real numbers a , b , and c ,

$$a(b + c) = ab + ac \quad \text{and} \quad (b + c)a = ba + ca.$$

F4. *Existence of Identity Elements* There exist two distinct real numbers, denoted 0 and 1, such that for every real number a ,

$$0 + a = a + 0 = a \quad \text{and} \quad 1 \cdot a = a \cdot 1 = a.$$

F5. *Existence of Additive Inverses* For every real number a , there is a real number, denoted $-a$ and called the **additive inverse** of a , such that

$$a + (-a) = (-a) + a = 0.$$

F6. *Existence of Reciprocals* For every real number $a \neq 0$, there is a real number, denoted $1/a$ or a^{-1} , called the **reciprocal** of a , such that

$$a \cdot \left(\frac{1}{a}\right) = \left(\frac{1}{a}\right) \cdot a = 1.$$



T1. *Cancellation Law for Addition* If $a + b = a + c$, then $b = c$. (In particular, this shows that the number 0 of Axiom F4 is unique.)

T2. *Possibility of Subtraction* Given a and b , there is exactly one x such that $a + x = b$. This x is denoted by $b - a$. In particular, $0 - a$ is the additive inverse of a , $-a$.

T3. $b - a = b + (-a)$.

T4. $-(-a) = a$.

T5. $a(b - c) = ab - ac$.

T6. $0 \cdot a = a \cdot 0 = 0$.

T7. *Cancellation Law for Multiplication* If $ab = ac$ and $a \neq 0$, then $b = c$. (In particular, this shows that the number 1 of Axiom F4 is unique.)

T8. *Possibility of Division* Given a and b with $a \neq 0$, there is exactly one x such that $ax = b$. This x is denoted by b/a and is called the **quotient** of b and a . In particular, $1/a$ is the reciprocal of a .

T9. If $a \neq 0$, then $b/a = b \cdot a^{-1}$.

T10. If $a \neq 0$, then $(a^{-1})^{-1} = a$.

T11. *Zero Product Property* If $ab = 0$, then $a = 0$ or $b = 0$.

T12. *Rule for Multiplication with Negative Signs*

$$(-a)b = a(-b) = -(ab), \quad (-a)(-b) = ab,$$

and

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}.$$

T13. *Equivalent Fractions Property*

$$\frac{a}{b} = \frac{ac}{bc}, \quad \text{if } b \neq 0 \text{ and } c \neq 0.$$

T14. *Rule for Addition of Fractions*

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \text{if } b \neq 0 \text{ and } d \neq 0.$$

T15. *Rule for Multiplication of Fractions*

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \text{if } b \neq 0 \text{ and } d \neq 0.$$

T16. *Rule for Division of Fractions*

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}, \quad \text{if } b \neq 0, c \neq 0, \text{ and } d \neq 0.$$

Appendix 2



Appendix 3

The real numbers also satisfy the following axioms, called the **order axioms**. It is assumed that among all real numbers there are certain ones, called the **positive real numbers**, that satisfy properties [Ord1](#), [Ord2](#), and [Ord3](#).

Ord1. For any real numbers a and b , if a and b are positive, so are $a + b$ and ab .

Ord2. For every real number $a \neq 0$, either a is positive or $-a$ is positive but not both.

Ord3. The number 0 is not positive.

The symbols $<$, $>$, $\leq$, and $\geq$, and negative numbers are defined in terms of positive numbers.

T17. *Trichotomy Law* For arbitrary real numbers a and b , exactly one of the three relations $a < b$, $b < a$, or $a = b$ holds.

T18. *Transitive Law* If $a < b$ and $b < c$, then $a < c$.

T19. If $a < b$, then $a + c < b + c$.

T20. If $a < b$ and $c > 0$, then $ac < bc$.

T21. If $a \neq 0$, then $a^2 > 0$.

T22. $1 > 0$.

T23. If $a < b$ and $c < 0$, then $ac > bc$.

T24. If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.

T25. If $ab > 0$, then both a and b are positive or both are negative.

T26. If $a < c$ and $b < d$, then $a + b < c + d$.

T27. If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.



Q5

Given $A = \{1, 3, 5, 7, 11, 13\}$ and $B = \{0, 2, 4, 6\}$, explain whether each of the following statements is true or false.

- (a) $\forall x \in B \forall y \in B (x - y \in B)$ False
- (b) $\forall x \in A \forall y \in A ((x < y) \wedge (y < 10) \rightarrow y - x \in B)$ True
- (c) $\forall x \in A \forall y \in B (x = y + 1)$ False
- (d) $\exists x \in A \exists y \in B (x = y + 1)$ True
- (e) $\exists y \in B \exists x \in A (x = y + 1)$ True

Note: (d) and (e) are equivalent.

Q5 Given $A = \{1,3,5,7,11,13\}$ and $B = \{0,2,4,6\}$, explain whether each of the following statements is true or false.

(f) $\forall x \in A \forall y \in B (x \neq y + 1)$ **False**

(g) $\exists x \in A \exists y \in B (x \neq y + 1)$ **True**

(h) $\forall x \in A \exists y \in B (x > y)$ **True**

(i) $\forall x \in A \exists y \in B (x \leq y)$ **False**

(j) $\exists y \in B \forall x \in A (x > y)$ **True**

Note: (f) is the negation of (d);
(g) is the negation of (c).

(c) $\forall x \in A \forall y \in B (x = y + 1)$

(d) $\exists x \in A \exists y \in B (x = y + 1)$

Q7 (a) Proof for: $\forall x \in \mathbb{R} (x^2 \geq 0)$. What is wrong with this “proof”?

There are 3 cases to consider: $x < 0$, $x = 0$ and $x > 0$.

If $x < 0$, for example, $x = -3$, then $x^2 = 9 \geq 0$;

if $x = 0$, then $x^2 = 0$;

if $x > 0$, say $x = 4$, then $x^2 = 16 \geq 0$.

Therefore, in all cases, $x^2 \geq 0$.

(b) Use the same logic in (a) to prove: $\forall x \in \mathbb{R} (x^3 = x)$.

Q7 (c) Proof for: $\forall x \in \mathbb{R} (x^2 \geq 0)$. What is wrong with this “proof”?

Prove by contradiction.

Suppose $x^2 < 0$ for all real numbers.

Let $x = 3$, then $x^2 = 9 > 0$ which is a contradiction.

Therefore, $\forall x \in \mathbb{R} (x^2 \geq 0)$.

(d) Use the same logic in (c) to prove: $\forall x \in \mathbb{R} (x^3 = x)$.

5-min Break

Q8 The following is a partial proof of the claim:

$$\forall x \in \mathbb{R} ((x^2 > x) \rightarrow (x < 0) \vee (x > 1)).$$

Appendix A, T25:
If $ab > 0$, then both a and b are positive or both are negative.

1. Let r be an arbitrarily chosen real number.
2. Suppose $r^2 > r$.
 - 2.1. Then $r^2 - r > 0$, or $r(r - 1) > 0$. (by basic algebra)
 - 2.2. So, both r and $r - 1$ are positive, or both are negative. (by Appendix A, T25)

(a) Do we need to include the case $r^2 \leq r$?

No. A conditional statement is true
(true by default/vacuously true) when
its hypotheses is false.

3. Therefore, $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow (x < 0) \vee (x > 1)).$

Q8 The following is a partial proof of the claim:

$$\forall x \in \mathbb{R} ((x^2 > x) \rightarrow (x < 0) \vee (x > 1)).$$

1. Let r be an arbitrarily chosen real number.
2. Suppose $r^2 > r$.
 - 2.1. Then $r^2 - r > 0$, or $r(r - 1) > 0$. (by basic algebra)
 - 2.2. So, both r and $r - 1$ are positive, or both are negative. (by Appendix A, T25)
 - 2.3. Case 1: both r and $r - 1$ are positive.
 - 2.3.1. $(r > 0) \wedge (r - 1 > 0) \rightarrow (r > 1)$.
 - 2.4. Case 2: both r and $r - 1$ are negative.
 - 2.4.1. $(r < 0) \wedge (r - 1 < 0) \rightarrow (r < 0)$.
 - 2.5. From lines 2.3.1 and 2.4.1, we have $(r > 1) \vee (r < 0)$.
 - 2.6. Therefore, $(r^2 > r) \rightarrow (r > 1) \vee (r < 0)$.
3. Therefore, $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow (x < 0) \vee (x > 1))$.

(b) Complete the proof.



Q8 The following is a partial proof of the claim:

$$\forall x \in \mathbb{R} \left((x^2 > x) \rightarrow (x < 0) \vee (x > 1) \right).$$

1. Let r be an arbitrarily chosen real number.
2. Suppose $r^2 > r$.
 - 2.1. Then $r^2 - r > 0$, or $r(r - 1) > 0$. (by basic algebra)
 - 2.2. So, both r and $r - 1$ are positive, or both are negative. (by Appendix A, T25)
3. Therefore, $\forall x \in \mathbb{R} \left((x^2 > x) \rightarrow (x < 0) \vee (x > 1) \right)$.

(c) Step 3 is an application of **universal generalization**. Explain what it means.

Q9 $T(v)$ = “ v took the Transformers ride”
 $G(v)$ = “ v took the Battlestar Galactica ride”
 $E(v)$ = “ v visited the Ancient Egypt”
 $W(v)$ = “ v watched the Water World show”

(a) Every visitor watched the Water World show.

$$\forall v \in V (W(v))$$

(b) Every visitor who took the Battlestar Galactica ride also took the Transformers ride.

$$\forall v \in V (G(v) \rightarrow T(v))$$

(c) There is a visitor who took both the Transformers ride and the Battlestar Galactica ride.

$$\exists v \in V (T(v) \wedge G(v))$$

Q9 (d) No visitor who visited the Ancient Egypt watched the Water World show.

$$\forall v \in V (E(v) \rightarrow \sim W(v))$$

Equivalently,
 $\sim$ (there is a visitor who visited the Ancient Egypt and watched the Water World show).

$$\begin{aligned} & \sim (\exists v \in V (E(v) \wedge W(v))) \\ \equiv & \forall v \in V \sim (E(v) \wedge W(v)) && \text{(negation of existential)} \\ \equiv & \forall v \in V (\sim E(v) \vee \sim W(v)) && \text{(De Morgan's law)} \\ \equiv & \forall v \in V (E(v) \rightarrow \sim W(v)) && \text{(implication law)} \end{aligned}$$

Q9

(e) Some visitors who took the Transformers ride also visited the Ancient Egypt but some (who took the Transformers ride) did not (visit the Ancient Egypt).

~~$$\exists v \in V (T(v) \wedge E(v)) \quad \wedge \quad \exists v \in V (T(v) \wedge \sim E(v))$$~~

Any issue?

$$\exists v \in V (T(v) \wedge E(v)) \quad \wedge \quad \exists u \in V (T(u) \wedge \sim E(u))$$

Q11 Prove that if n is a product of two positive integers a and b , then $a \leq n^{1/2}$ or $b \leq n^{1/2}$.

Proof by contraposition:

$$p \rightarrow q \equiv \sim q \rightarrow \sim p$$

1. If $a > n^{1/2}$ and $b > n^{1/2}$, then n is not a product of a and b .
($\sim(a \leq n^{1/2} \vee b \leq n^{1/2}) \equiv (a > n^{1/2} \wedge b > n^{1/2})$ by De Morgan's law)
2. Suppose $a > n^{1/2}$ and $b > n^{1/2}$, then $ab > n^{1/2} \cdot n^{1/2} = n$. (by Appendix A, T27)
3. Since $ab \neq n$, the contrapositive statement is true.
4. Therefore, the original statement is true.



Appendix A, T27:

$\forall a, b, c \in \mathbb{R}$, if $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

Q11 Prove that if n is a product of two positive integers a and b , then $a \leq n^{1/2}$ or $b \leq n^{1/2}$.

Proof by contradiction:

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

1. Suppose not (i.e., taking the negation of the given statement), we have $n = ab$ and $(a > n^{1/2} \text{ and } b > n^{1/2})$. (by De Morgan's law)
2. Since $a > n^{1/2}$ and $b > n^{1/2}$, we have $ab > n^{1/2} \cdot n^{1/2} = n$. (by Appendix A, T27)
3. This contradicts $n = ab$.
4. Therefore, the original statement is true.



Appendix A, T27:

$\forall a, b, c \in \mathbb{R}$, if $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

Q11 Prove that if n is a product of two positive integers a and b , then $a \leq n^{1/2}$ or $b \leq n^{1/2}$.

Other proofs welcome!

Q11 Prove that if n is a product of two positive integers a and b , then $a \leq n^{1/2}$ or $b \leq n^{1/2}$.

Note: For $x, y \in \mathbb{R}$, we define x^y s.t. $x > 0 \wedge y > 0 \rightarrow x^y > 0$.

e.g. $4^{1/2} = 2$ (does not include -2).

We do not include the 'negative result' into consideration as we prefer for the function x^y to be continuous when x is a positive real number.

Otherwise, counterexamples of said statement exist.

- Q10
1. If an obj is above all the triangles, then it is above all the blue obj.
 2. If an obj is not above all the gray obj, then it is not a square.
 3. Every black obj is a square.
 4. Every obj that is above all the gray obj is above all the triangles.
 - $\therefore$ If an obj is black, then it is above all the blue obj.

(a) Reorder the premises to show that the conclusion follows as a valid consequence from the premises by applying universal transitivity.

3. If an obj is black, then it is a square.
2. If an obj is a square, then it is above all the gray obj.
4. If an obj is above all the gray obj, then it is above all the triangles.
1. If an obj is above all the triangles, then it is above all the blue obj.
- $\therefore$ If an obj is black, then it is above all the blue obj.

Q10

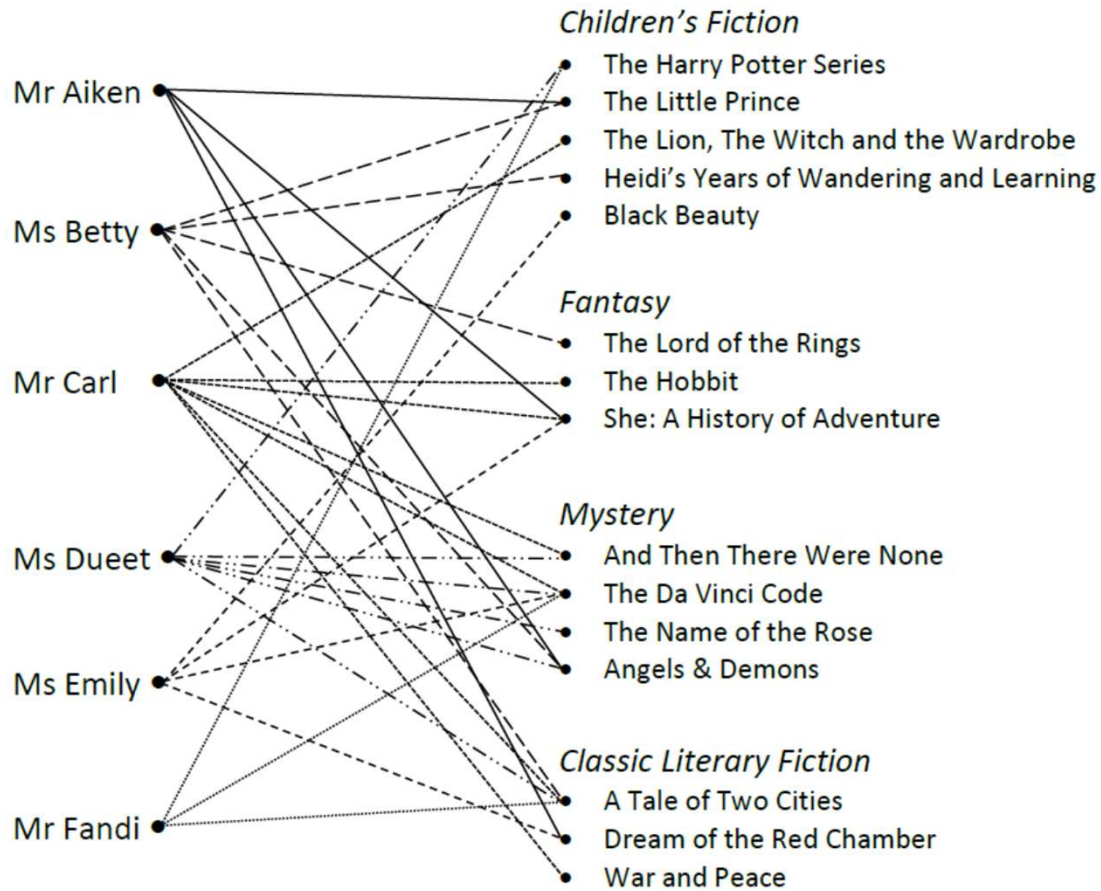
3. If an obj is black, then it is a square.
2. If an obj is a square, then it is above all the gray obj.
4. If an obj is above all the gray obj, then it is above all the triangles.
1. If an obj is above all the triangles, then it is above all the blue obj.
- $\therefore$ If an obj is black, then it is above all the blue obj.

(b) Rewrite your answer in (a) using predicates and quantified statements.

Let O , the domain of discourse, be the set of objects.

3. $\forall x \in O (Black(x) \rightarrow Square(x))$.
2. $\forall x \in O (Square(x) \rightarrow (\forall y \in O (Gray(y) \rightarrow Above(x, y))))$.
4. $\forall x \in O ((\forall y \in O (Gray(y) \rightarrow Above(x, y))) \rightarrow (\forall z \in O (Triangle(z) \rightarrow Above(x, z))))$.
1. $\forall x \in O ((\forall z \in O (Triangle(z) \rightarrow Above(x, z))) \rightarrow (\forall w \in O (Blue(w) \rightarrow Above(x, w))))$.
- $\therefore \forall x \in O (Black(x) \rightarrow (\forall w \in O (Blue(w) \rightarrow Above(x, w))))$.

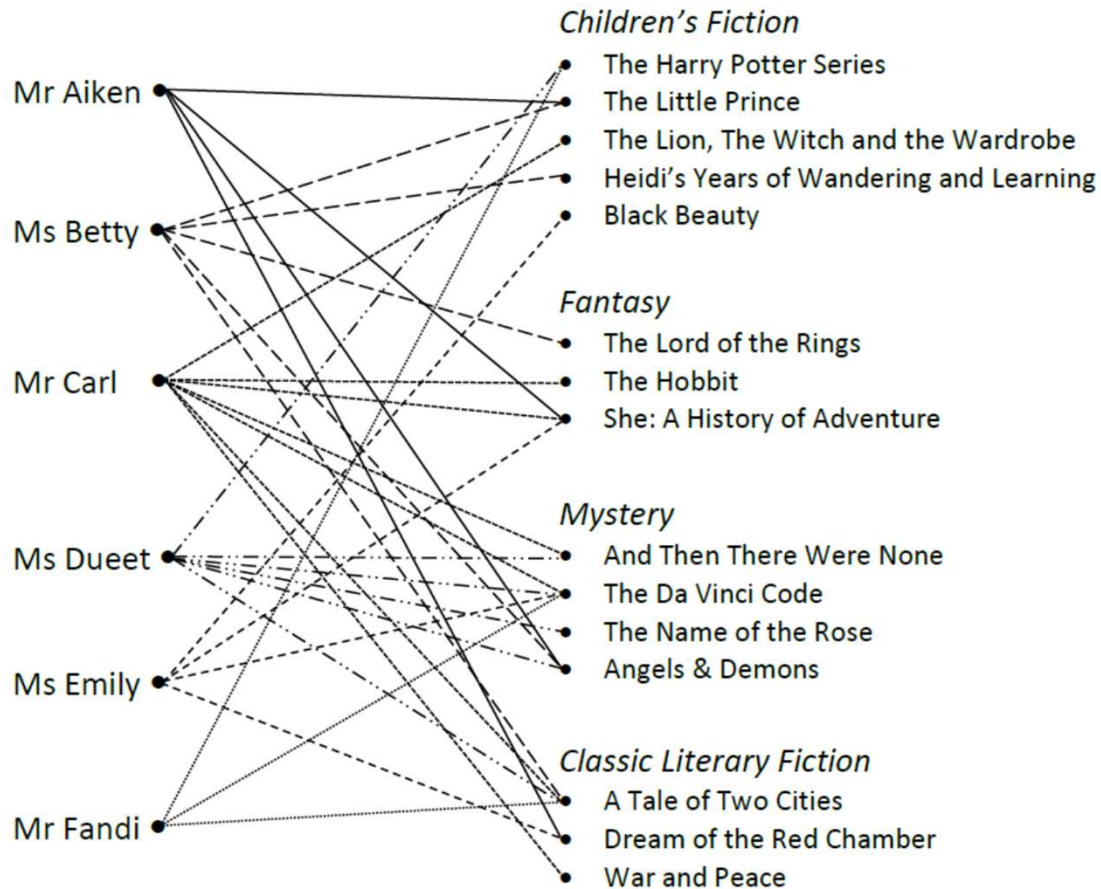
Q6



(a) Some title is read by all the female readers.

False;
None of the titles is read by all 3 female readers.

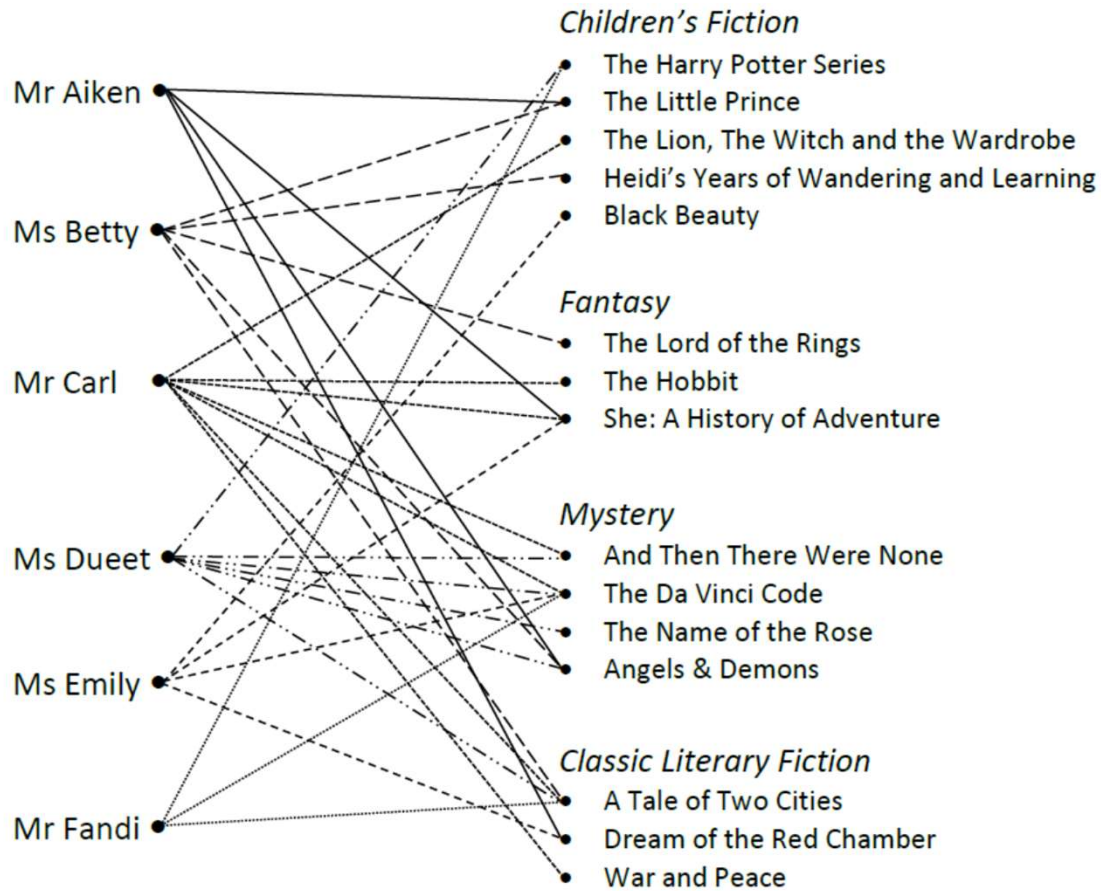
Q6



(b) Every reader reads some title in every genre.

False;
Ms Dueet doesn't read any Fantasy title.

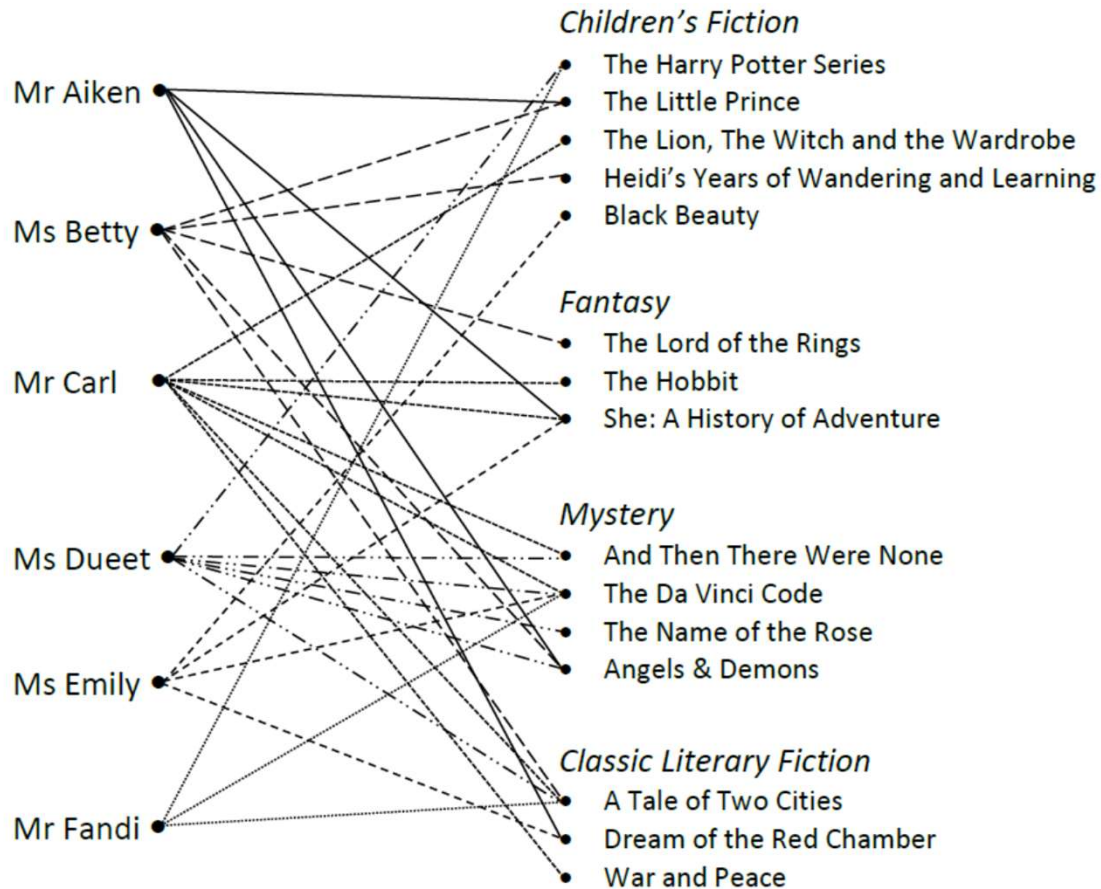
Q6



(c) Some reader reads all titles of some genre.

True;
Ms Dueet reads all Mystery titles.

Q6



(d) There is some genre for which some reader does not read any of its titles.

True;
None of the Fantasy titles is read by Ms Dueet/Mr Fandi.

THE END